API response times

https://changecarapi.wowcar.co.th · measured 16 October 2026

Every GET endpoint of the ChangeCars API (plus login) was called once, one request at a time, each on a new HTTPS connection. Write requests (POST, PATCH, DELETE) were not called because they change live data. **Total** is what the app waits: connection, SSL, server work and download. **Server** is the time from sending the request to the first byte of the answer; it includes one network round trip.

Requests measured	Median total	Median server	Slower than 3 s
155	**1.91 s**	**0.96 s**	**16**

Response time	APIs	Share
Under 2 seconds	88	57%
2 to 3 seconds	51	33%
Over 3 seconds	16	10%

Why the app feels slow

- **One endpoint is very large.** `/vehicles/filters/all` returns 40 MB (7.6 MB compressed) and took 7.8 to 12.6 seconds. On mobile data this can take much longer. If the app calls it on start-up or on the filter screen, the app waits for it.
- **The server runs on shared hosting** (Plesk with Phusion Passenger). Even the empty health check spends about 0.35 to 0.8 seconds on the server. Passenger also stops idle apps, so the first request after a quiet period waits for the app to start again.
- **Every new connection costs about 0.9 seconds** before any data moves (0.43 s to connect plus 0.44 s for SSL, measured from Pakistan through Cloudflare's Johannesburg edge). Users in South Africa will see less of this.
- Compression (gzip) and Cloudflare caching are already working for the list and filter endpoints.

Recommendations

- Stop calling `/vehicles/filters/all` from the app. Use the small filter endpoints (`/vehicles/filters/colours`, `/makes`, `/models`, ...; a few KB each) or download the compressed snapshot once via `/vehicle-filters/manifest` and keep it on the phone until it changes.
- Move the API to a VPS or a container host where the app stays running, or at least disable Passenger's idle shutdown.
- Reuse one connection in the app (HTTP keep-alive / HTTP/2) and request the APIs of a screen in parallel instead of one after another.
- Cache `/vehicle-filters/manifest` (4-5 s) and look at `/admin/console/settings` (2.9 s on the server for a read-only page).

Slowest 15 requests

#	API	Total	Server	Size
1	GET /vehicles/filters/all	**12.64 s**	5.28 s	40.8 MB
2	GET /vehicles/filters/all	**7.80 s**	0.44 s	40.8 MB
3	GET /vehicle-filters/manifest	**5.15 s**	4.28 s	0.4 KB
4	GET /vehicle-filters/manifest	**4.76 s**	3.89 s	0.4 KB

#	API	Total	Server	Size
5	GET /web/dashboard/dealers	**4.63 s**	1.95 s	893.9 KB
6	GET /web/cars/all	**4.39 s**	1.22 s	2.2 MB
7	GET /vehicle-filters/snapshots/inv-914bb1330c2729ca134eeb5251a68dcb99feb1fc8f5c800be36bc4af98762fcc.json.gz	**4.34 s**	0.88 s	3.6 MB
8	GET /vehicle-filters/snapshots/inv-914bb1330c2729ca134eeb5251a68dcb99feb1fc8f5c800be36bc4af98762fcc.json.gz	**4.00 s**	0.44 s	3.6 MB
9	GET /seo/sitemap-vehicles-1.xml	**3.89 s**	1.02 s	1.4 MB
10	GET /admin/console/settings	**3.79 s**	2.92 s	0.2 KB
11	GET /vehicles/facets	**3.33 s**	0.84 s	689.8 KB
12	GET /vehicles/filters/variants	**3.32 s**	1.05 s	535.0 KB
13	GET /vehicles/facets	**3.30 s**	0.82 s	689.8 KB
14	GET /seo/sitemap-static.xml	**3.27 s**	1.62 s	164.6 KB
15	GET /vehicles/filters/variants	**3.18 s**	0.79 s	535.0 KB

All requests (slowest first)

Colour of the total time: red over 3 s, amber 2-3 s, green under 2 s. Two requests returned 404 because the sample record does not exist in the data (a customer document and a bidding session the demo dealer is not invited to).

#	API	Status	Total	Server	Size
1	GET /vehicles/filters/all	200	**12.64 s**	5.28 s	40.8 MB
2	GET /vehicles/filters/all	200	**7.80 s**	0.44 s	40.8 MB
3	GET /vehicle-filters/manifest	200	**5.15 s**	4.28 s	0.4 KB
4	GET /vehicle-filters/manifest	200	**4.76 s**	3.89 s	0.4 KB
5	GET /web/dashboard/dealers	200	**4.63 s**	1.95 s	893.9 KB
6	GET /web/cars/all	200	**4.39 s**	1.22 s	2.2 MB
7	GET /vehicle-filters/snapshots/inv-914bb1330c2729ca134eeb5251a68dcb99feb1fc8f5c800be36bc4af98762fcc.json.gz	200	**4.34 s**	0.88 s	3.6 MB
8	GET /vehicle-filters/snapshots/inv-914bb1330c2729ca134eeb5251a68dcb99feb1fc8f5c800be36bc4af98762fcc.json.gz	200	**4.00 s**	0.44 s	3.6 MB
9	GET /seo/sitemap-vehicles-1.xml	200	**3.89 s**	1.02 s	1.4 MB
10	GET /admin/console/settings	200	**3.79 s**	2.92 s	0.2 KB
11	GET /vehicles/facets	200	**3.33 s**	0.84 s	689.8 KB
12	GET /vehicles/filters/variants	200	**3.32 s**	1.05 s	535.0 KB
13	GET /vehicles/facets	200	**3.30 s**	0.82 s	689.8 KB
14	GET /seo/sitemap-static.xml	200	**3.27 s**	1.62 s	164.6 KB
15	GET /vehicles/filters/variants	200	**3.18 s**	0.79 s	535.0 KB
16	GET /me/documents	200	**3.09 s**	1.27 s	0.5 KB
17	GET /vehicles/specials	200	**2.99 s**	2.00 s	34.6 KB
18	GET /admin/vehicles	200	**2.92 s**	1.46 s	84.8 KB
19	GET /seo/vehicles/2026-haval-jolion-pro-1-5t-premium-dct-fbook-1533319/structured-data	200	**2.76 s**	1.83 s	0.9 KB
20	GET /vehicles/bd0ac17e-250a-5430-a354-340933904e27/dealer-vehicles	200	**2.76 s**	0.93 s	10.5 KB
21	GET /vehicles/bd0ac17e-250a-5430-a354-340933904e27/market-price	200	**2.70 s**	1.35 s	0.1 KB
22	GET /vehicles/bd0ac17e-250a-5430-a354-340933904e27/similar	200	**2.58 s**	1.71 s	10.8 KB
23	GET /admin/vehicles/bd0ac17e-250a-5430-a354-340933904e27	200	**2.49 s**	1.56 s	19.5 KB
24	GET /admin/console/snapshots	200	**2.46 s**	1.59 s	1.2 KB
25	GET /vehicles	200	**2.45 s**	1.50 s	34.4 KB
26	GET /web/dashboard/dealers-by-id	200	**2.44 s**	1.54 s	0.0 KB
27	GET /vehicles/2026-haval-jolion-pro-1-5t-premium-dct-fbook-1533319	200	**2.43 s**	1.50 s	32.1 KB
28	GET /admin/system/outbox	200	**2.37 s**	1.45 s	0.1 KB
29	GET /dealer/dashboard	200	**2.37 s**	1.45 s	0.5 KB
30	GET /vehicles/hot-sellers	200	**2.35 s**	1.38 s	36.4 KB
31	GET /vehicles/filters/fuel-types	200	**2.34 s**	1.38 s	0.3 KB
32	GET /me/sell-requests/	200	**2.34 s**	1.30 s	0.1 KB
33	GET /web/cars/bd0ac17e250a5430a354340933904e27/dealer-cars	200	**2.34 s**	1.43 s	15.2 KB

#	API	Status	Total	Server	Size
34	GET /vehicles/filters/colours	200	2.34 s	1.43 s	2.9 KB
35	GET /me/documents//view	404	2.30 s	1.42 s	0.2 KB
36	GET /web/cars/count	200	2.30 s	1.32 s	0.1 KB
37	GET /content/faqs	200	2.29 s	1.37 s	1.4 KB
38	GET /web/cars/bd0ac17e250a5430a354340933904e27	200	2.29 s	1.29 s	32.9 KB
39	GET /vehicles/filters/transmissions	200	2.27 s	1.39 s	0.1 KB
40	GET /vehicles/compare	200	2.27 s	1.33 s	5.1 KB
41	GET /dealer/documents	200	2.26 s	1.39 s	0.0 KB
42	GET /dealer/vehicles/01a11d8e-4d1f-7512-9000-d8f60954c991/images	200	2.26 s	1.37 s	0.0 KB
43	GET /web/dashboard/inventory	200	2.25 s	1.38 s	0.0 KB
44	GET /me/enquiries/	200	2.25 s	1.39 s	0.1 KB
45	GET /vehicles/filters/colours	200	2.25 s	1.37 s	0.8 KB
46	GET /web/cars/bd0ac17e250a5430a354340933904e27/market-price	200	2.25 s	1.33 s	0.0 KB
47	GET /dealer/enquiries/	200	2.25 s	1.39 s	0.1 KB
48	GET /admin/users	200	2.24 s	1.38 s	3.1 KB
49	GET /dealer/profile	200	2.24 s	1.37 s	1.2 KB
50	GET /vehicles/filters/drives	200	2.23 s	1.35 s	0.2 KB
51	GET /me/favourites/ids	200	2.22 s	1.34 s	0.0 KB
52	GET /dealer/bidding/sessions/01a12032-0616-759a-a02f-2fab896c2a4b	404	2.21 s	1.33 s	0.3 KB
53	GET /dealer/vehicles/01a11d8e-4d1f-7512-9000-d8f60954c991	200	2.21 s	1.32 s	4.4 KB
54	GET /content/media/	200	2.20 s	1.34 s	0.1 KB
55	GET /admin/faqs	200	2.20 s	1.33 s	1.4 KB
56	GET /catalogue/variants/compare	200	2.19 s	1.33 s	3.1 KB
57	GET /admin/dealers	200	2.18 s	1.27 s	29.5 KB
58	GET /auth/me	200	2.18 s	1.31 s	0.2 KB
59	GET /content/articles	200	2.16 s	1.21 s	0.7 KB
60	GET /dealer/staff/01a11d8e-4c3d-705f-aac4-1220d3e317fc/activity	200	2.15 s	1.19 s	1.0 KB
61	GET /admin/enquiries/	200	2.15 s	1.28 s	0.1 KB
62	GET /admin/dealers/01a11d8e-4c26-76ac-8182-66e8e7197417	200	2.15 s	1.27 s	2.4 KB
63	GET /dealer/leads	200	2.14 s	1.26 s	0.1 KB
64	GET /seo/sitemap.xml	200	2.14 s	1.25 s	0.6 KB
65	GET /me	200	2.10 s	1.23 s	0.4 KB
66	GET /catalogue/variants/00201de7-74ab-5387-abe0-ace393c4d2da	200	2.08 s	1.15 s	1.6 KB
67	POST /auth/login	200	2.07 s	1.20 s	0.8 KB
68	GET /web/cars/premium	200	1.99 s	0.95 s	29.0 KB
69	GET /web/articles	200	1.96 s	1.04 s	0.0 KB
70	GET /dealer/vehicles	200	1.96 s	0.99 s	16.1 KB

#	API	Status	Total	Server	Size
71	GET /web/cars	200	**1.96 s**	0.98 s	47.1 KB
72	GET /web/cars/bd0ac17e250a5430a354340933904e27/similar	200	**1.95 s**	1.05 s	10.5 KB
73	GET /promotions/	200	**1.93 s**	1.03 s	0.0 KB
74	GET /vehicles/filters/colours	200	**1.93 s**	1.07 s	0.5 KB
75	GET /dealers/sandton-auto	200	**1.92 s**	1.05 s	0.8 KB
76	GET /me/sessions	200	**1.92 s**	1.02 s	1.1 KB
77	GET /admin/console/imports	200	**1.91 s**	1.03 s	7.2 KB
78	GET /me/favourites	200	**1.91 s**	0.98 s	0.1 KB
79	GET /admin/articles/01a11d8e-4d32-7065-be75-8a172a724281	200	**1.90 s**	0.96 s	0.8 KB
80	GET /admin/console/quality	200	**1.89 s**	0.96 s	20.3 KB
81	GET /web/dashboard/overview	200	**1.89 s**	1.00 s	0.5 KB
82	GET /me/sell-requests	200	**1.89 s**	1.01 s	0.1 KB
83	GET /web/articles/how-to-sell-your-car-for-the-best-price	200	**1.88 s**	0.96 s	0.6 KB
84	GET /me/recently-viewed	200	**1.88 s**	0.96 s	0.0 KB
85	GET /catalogue/features	200	**1.88 s**	1.01 s	2.8 KB
86	GET /vehicles/count	200	**1.87 s**	0.92 s	0.1 KB
87	GET /web/article-categories	200	**1.87 s**	0.94 s	0.8 KB
88	GET /admin/enquiries	200	**1.87 s**	0.90 s	0.1 KB
89	GET /web/cars/featured	200	**1.86 s**	0.96 s	13.9 KB
90	GET /admin/catalogue/models/01a11d97-edeb-73a8-b042-de9439c8a54b/variants	200	**1.86 s**	0.97 s	3.4 KB
91	GET /vehicles/filters/models	200	**1.85 s**	0.85 s	56.5 KB
92	GET /web/articles/featured	200	**1.85 s**	0.94 s	0.6 KB
93	GET /web/cars/bd0ac17e250a5430a354340933904e27/popular-dealers	200	**1.85 s**	0.88 s	1.2 KB
94	GET /web/dashboard/staff	200	**1.84 s**	0.95 s	0.2 KB
95	GET /dealer/leads/	200	**1.84 s**	0.96 s	0.1 KB
96	GET /admin/catalogue/makes/01a11d8e-4873-7357-9f03-e46318778f7b/models	200	**1.84 s**	0.95 s	10.1 KB
97	GET /web/articles/latest	200	**1.83 s**	0.93 s	0.0 KB
98	GET /catalogue/categories	200	**1.83 s**	0.96 s	5.4 KB
99	GET /dealers	200	**1.83 s**	0.91 s	19.7 KB
100	GET /web/dashboard/dealers/01a11d8e-4c26-76ac-8182-66e8e7197417	200	**1.82 s**	0.89 s	0.9 KB
101	GET /admin/articles	200	**1.82 s**	0.94 s	0.6 KB
102	GET /content/media	200	**1.82 s**	0.95 s	0.1 KB
103	GET /admin/bidding-sessions	200	**1.82 s**	0.93 s	0.1 KB
104	GET /catalogue/makes/toyota/models	200	**1.81 s**	0.94 s	3.9 KB
105	GET /admin/system/outbox/events	200	**1.81 s**	0.95 s	0.1 KB
106	GET /catalogue/makes	200	**1.81 s**	0.92 s	9.4 KB
107	GET /vehicles/filters/makes	200	**1.81 s**	0.94 s	4.0 KB

#	API	Status	Total	Server	Size
108	GET /dealer/leads/stats	200	1.81 s	0.94 s	0.1 KB
109	GET /me/dashboard	200	1.81 s	0.94 s	0.6 KB
110	GET /admin/console/leads/	200	1.81 s	0.91 s	0.1 KB
111	GET /web/cars/recent	200	1.81 s	0.91 s	13.3 KB
112	GET /admin/media	200	1.80 s	0.92 s	0.1 KB
113	GET /web/dashboard/activity	200	1.80 s	0.88 s	1.4 KB
114	GET /admin/catalogue/makes	200	1.80 s	0.88 s	21.8 KB
115	GET /me/notifications/unread-count	200	1.79 s	0.91 s	0.0 KB
116	GET /admin/console/leads	200	1.79 s	0.92 s	0.1 KB
117	GET /me/notifications	200	1.79 s	0.91 s	0.1 KB
118	GET /web/dashboard/leads	200	1.78 s	0.89 s	0.0 KB
119	GET /dealer/bidding/sessions	200	1.78 s	0.85 s	0.1 KB
120	GET /vehicles/filters/models	200	1.78 s	0.79 s	56.5 KB
121	GET /admin/audit-logs	200	1.78 s	0.91 s	7.5 KB
122	GET /admin/pages	200	1.77 s	0.91 s	1.2 KB
123	GET /admin/stats	200	1.77 s	0.91 s	1.1 KB
124	GET /dealer/offers	200	1.77 s	0.91 s	0.1 KB
125	GET /admin/sell-requests	200	1.77 s	0.90 s	0.1 KB
126	GET /admin/newsletter/subscribers.csv	200	1.77 s	0.91 s	0.1 KB
127	GET /admin/users/01a11d8e-4c1d-767f-8844-cee453be2fd9	200	1.76 s	0.91 s	0.7 KB
128	GET /admin/sell-requests/	200	1.76 s	0.90 s	0.1 KB
129	GET /me/enquiries	200	1.76 s	0.89 s	0.1 KB
130	GET /admin/promotions	200	1.75 s	0.88 s	0.0 KB
131	GET /admin/catalogue/categories	200	1.75 s	0.87 s	5.4 KB
132	GET /admin/vehicles/review-queue	200	1.75 s	0.89 s	0.1 KB
133	GET /me/saved-searches	200	1.74 s	0.85 s	0.0 KB
134	GET /dealer/me	200	1.74 s	0.88 s	0.5 KB
135	GET /me/notifications/preferences	200	1.74 s	0.83 s	1.8 KB
136	GET /content/article-categories	200	1.74 s	0.88 s	2.1 KB
137	GET /catalogue/makes/toyota/models/86	200	1.74 s	0.88 s	3.8 KB
138	GET /admin/newsletter/subscribers	200	1.73 s	0.86 s	0.1 KB
139	GET /dealer/staff	200	1.72 s	0.86 s	1.6 KB
140	GET /content/articles/how-to-sell-your-car-for-the-best-price	200	1.72 s	0.85 s	0.9 KB
141	GET /dealer/branches	200	1.72 s	0.82 s	0.5 KB
142	GET /dealer/enquiries	200	1.71 s	0.85 s	0.1 KB
143	GET /admin/pages/01a11d8e-4954-70dd-9c09-6f329f79749a	200	1.71 s	0.85 s	0.4 KB
144	GET /me/export	200	1.71 s	0.84 s	0.7 KB

#	API	Status	Total	Server	Size
145	GET /me/offers	200	**1.71 s**	0.84 s	0.1 KB
146	GET /promotions	200	**1.70 s**	0.83 s	0.0 KB
147	GET /content/pages/privacy-policy	200	**1.70 s**	0.84 s	0.4 KB
148	GET /vehicles/filters/transmissions	200	**1.70 s**	0.79 s	0.1 KB
149	GET /vehicles/filters/makes	200	**1.69 s**	0.80 s	4.0 KB
150	GET /health/ready	200	**1.68 s**	0.78 s	0.1 KB
151	GET /health/live	200	**1.68 s**	0.79 s	0.0 KB
152	GET /vehicles/filters/fuel-types	200	**1.66 s**	0.80 s	0.3 KB
153	GET /vehicles/filters/drives	200	**1.65 s**	0.78 s	0.2 KB
154	GET /vehicles/filters/colours	200	**1.34 s**	0.46 s	2.9 KB
155	GET /vehicles/filters/colours	200	**1.31 s**	0.44 s	2.9 KB